Multiple Qubits: Tensor Products, CNOT, Bit Ordering, and Entanglement

EL4043 – Teknologi dan Komputasi Kuantum

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September 23, 2026

Learning Objectives

Our topics of discussion:

1. The state space grows exponentially with the number of qubits.
2. Multi-qubit states using the tensor product.
3. Computational basis of a multi-qubit system.
4. Apply local operations to individual qubits.
5. Understand the *CNOT* gate and its action on two qubits.
6. Understand Qiskit's qubit and classical-bit ordering.
7. Explain the basic idea of quantum entanglement.

Main idea

Multiple qubits are not simply "many independent qubits". The tensor-product structure allows quantum systems to exhibit correlations that have no classical equivalent.

From One Qubit to Multiple Qubits

A single qubit has state

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,,$$

where

$$|\alpha|^2 + |\beta|^2 = 1.$$

The state vector belongs to a two-dimensional Hilbert space:

$$\mathcal{H}_1 \cong \mathbb{C}^2.$$

For two qubits,

$$\mathcal{H}_2 = \mathcal{H}_1 \otimes \mathcal{H}_1 = \mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4.$$

For n qubits,

$$\boxed{\mathcal{H}_n = (\mathbb{C}^2)^{\otimes n} \cong \mathbb{C}^{2^n}}$$

How Fast Does the State Space Grow?

Number of qubits	Dimension
1	2
2	4
3	8
4	16
5	32
10	1024
20	1 048 576
50	$2^{50} \approx 1.13 \times 10^{15}$
100	$2^{100} \approx 1.27 \times 10^{30}$

Key point

The exponential Hilbert-space dimension is one of the central reasons quantum systems become difficult to simulate classically.

Tensor Product

Suppose qubit A is in

$$|\psi\rangle_A = \alpha|0\rangle + \beta|1\rangle$$

and qubit B is in

$$|\phi\rangle_B = \gamma|0\rangle + \delta|1\rangle\,.$$

The combined state is

$$\boxed{|\psi\rangle_{AB} = |\psi\rangle_A \otimes |\phi\rangle_B}$$

Expanding,

$$\begin{aligned}|\psi\rangle_{AB} &= (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle) \\ &= \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle\,.\end{aligned}$$

Thus a two-qubit state has four computational-basis components.

Tensor Product of Basis States

The computational basis for two qubits is

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

Using column vectors,

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

For example,

$$|01\rangle = |0\rangle \otimes |1\rangle.$$

Therefore,

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

Similarly,

$$|10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

Worked Example: $|+0\rangle$

Recall

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Consider

$$|+0\rangle = |+\rangle \otimes |0\rangle .$$

Therefore,

$$|+0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle$$
$$= \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle).$$

In vector form,

$$|+0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

This is a **product state**.

Exercise: Expand the Tensor Product

Consider

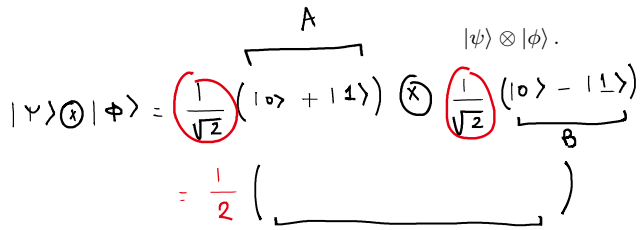
$$\rightarrow \quad |\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

and

$$\rightarrow \quad |\phi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

Find

$|\psi\rangle \otimes |\phi\rangle$.



Exercise: Expand the Tensor Product

Consider

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

and

$$|\phi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

Find

$$|\psi\rangle \otimes |\phi\rangle .$$

$$|\psi\rangle \otimes |\phi\rangle = \frac{1}{2}(|0\rangle + |1\rangle) \otimes (|0\rangle - |1\rangle)$$
$$= \boxed{\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)}.$$

Computational Basis for n Qubits

For one qubit:

$$\{|0\rangle, |1\rangle\}.$$

For two qubits:

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}.$$

For three qubits:

$$\{|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle\}.$$

In general,

$$|x_{n-1}\cdots x_1 x_0\rangle, \qquad x_i \in \{0,1\}$$

and there are

$$2^n$$

n: number of qubits

computational-basis states.

Measurement in the Computational Basis

Consider

$$|\psi\rangle = \alpha\,|00\rangle + \beta\,|01\rangle + \gamma\,|10\rangle + \delta\,|11\rangle\,.$$

When measured in the computational basis:

Outcome	Probability
00	$\lvert\alpha\rvert^2$
01	$\lvert\beta\rvert^2$
10	$\lvert\gamma\rvert^2$
11	$\lvert\delta\rvert^2$

with

$$|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1.$$

Remember

The amplitudes may be complex. Measurement probabilities are obtained from their squared magnitudes.

Quick Exercise

Suppose

$$|\psi\rangle = \frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle + |11\rangle\right).$$

What is the probability of measuring $|10\rangle$?

Quick Exercise

Suppose

$$|\psi\rangle = \frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle + |11\rangle\right).$$

What is the probability of measuring

$$|10\rangle?$$

The amplitude of $|10\rangle$ is

$$\frac{1}{2}.$$

Therefore,

$$\boxed{P(10) = \left|\frac{1}{2}\right|^2 = \frac{1}{4}}$$

So each of the four computational-basis outcomes occurs with probability 25%.

Local Operations

A **local operation** acts on only one qubit.
For example, for two qubits:

$$X \otimes I$$

applies X to the first qubit and does nothing to the second.
Similarly,

$$I \otimes X$$

does nothing to the first qubit and applies X to the second.

Recall

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Therefore,

$$\boxed{X \otimes I}$$

is a 4×4 matrix acting on the two-qubit Hilbert space.

Example of a Local Operation

Start with

$$|00\rangle .$$

Apply X to the first qubit:

$$(X \otimes I)\,|00\rangle .$$

Since

$$X\,|0\rangle = |1\rangle$$

and

$$I\,|0\rangle = |0\rangle ,$$

we obtain

$$\boxed{(X \otimes I)\,|00\rangle = |10\rangle}$$

Similarly,

$$\boxed{(I \otimes X)\,|00\rangle = |01\rangle}$$

Important

Local gates can change individual qubits, but they do not by themselves create entanglement from an initially separable state.

Matrix Representation of $X \otimes I$

Using the Kronecker product,

$$X \otimes I = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Therefore,

$$\boxed{X \otimes I = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}}$$

and

$$(X \otimes I) \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

Hence

$$|00\rangle \longrightarrow |10\rangle.$$

The CNOT Gate

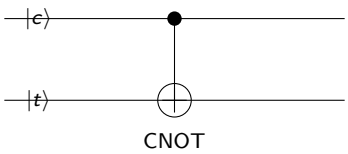
The Controlled-NOT gate is a fundamental two-qubit gate.
It has:

- one **control** qubit,
- one **target** qubit.

The target qubit is flipped only when the control qubit is 1.

$$\boxed{\mathrm{CNOT}\,|c, t\rangle = |c, t \oplus c\rangle}$$

where $\oplus$ denotes XOR.



CNOT Truth Table

$$\mathrm{CNOT}\,|c,t\rangle = |c, c \oplus t\rangle$$

Input	Operation	Output
$\|00\rangle$	target unchanged	$\|00\rangle$
$\|01\rangle$	target unchanged	$\|01\rangle$
$\|10\rangle$	target flips	$\|11\rangle$
$\|11\rangle$	target flips	$\|10\rangle$

Key idea

The control qubit itself is not changed by the CNOT.

CNOT Matrix

Using the computational basis ordering

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\},$$

the CNOT matrix is

$$\boxed{U_{\mathrm{CNOT}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}}$$

For example,

$$U_{\mathrm{CNOT}} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Therefore,

$$\boxed{|10\rangle \longrightarrow |11\rangle}$$

Superposition + CNOT

Start with

$$|00\rangle\,.$$

Apply a Hadamard gate to the first qubit:

$$(H \otimes I)\,|00\rangle\,.$$

Since

$$H\,|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle),$$

we obtain

$$\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle).$$

Now apply CNOT:

$$\mathrm{CNOT}\,\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle).$$

Using

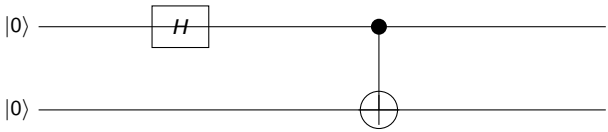
$$|00\rangle \to |00\rangle\,, \qquad |10\rangle \to |11\rangle\,,$$

we get

$$\boxed{|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)}$$

This is a Bell state.

Circuit for Creating a Bell State



$$|00\rangle \xrightarrow{H\otimes I} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{\text{CNOT}} \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Why Is CNOT Different from a Local Gate?

Compare

$$X \otimes I$$

with

$$\text{CNOT}.$$

A local gate acts independently on one subsystem:

$$|\psi\rangle_A \otimes |\phi\rangle_B \to (U|\psi\rangle_A) \otimes |\phi\rangle_B .$$

A two-qubit gate can correlate the two subsystems.
For example,

$$\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

The output cannot be written as

$$|\psi\rangle_A \otimes |\phi\rangle_B .$$

Therefore the output is entangled.

Qiskit Qubit Ordering

In Qiskit, qubits are indexed as

$$q_0, q_1, \ldots, q_{n-1}.$$

A circuit is usually displayed with q_0 at the top. However, for integer interpretation,

$$\boxed{q_0 \text{ is the least-significant bit}}$$

and

$$q_{n-1}$$

is the most-significant bit. For two qubits:

$$\boxed{q_1 q_0}$$

is the usual bit-string interpretation. Therefore:

State	Integer
$\|00\rangle$	0
$\|01\rangle$	1
$\|10\rangle$	2
$\|11\rangle$	3

Qiskit Example: Apply X to q_0

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(2)

qc.x(0)

print(qc)
```

The circuit performs

$$|00\rangle \xrightarrow{X_{q_0}} |01\rangle$$

when the computational state is written as

$$q_1 q_0.$$

Therefore the displayed bit string is

$$\boxed{01}.$$

Common source of confusion

The top wire is q_0, but q_0 is the rightmost bit when interpreting a computational-basis string as an integer.

Qiskit Example: CNOT

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(2)

qc.x(0)
qc.cx(0, 1)

print(qc)
```

Step 1:

$$|00\rangle \xrightarrow{X_{q_0}} |01\rangle .$$

Step 2:
$q_0 = 1$, so the CNOT flips q_1:

$$|01\rangle \longrightarrow |11\rangle .$$

Therefore,

$$\boxed{|00\rangle \rightarrow |01\rangle \rightarrow |11\rangle}$$

in Qiskit's computational-bit-string convention.

Mathematical Ordering vs Qiskit Ordering

Suppose we have two qubits:

$$|q_1 q_0\rangle .$$

The integer represented by the bit string is

$$q_1 2^1 + q_0 2^0.$$

Therefore,

$$|10\rangle$$

corresponds to integer 2. In Qiskit:

$$q_0 = \text{least significant bit}, \qquad q_1 = \text{most significant bit}.$$

Practical rule

Always check whether you are talking about:

1. circuit-wire order,
2. tensor-product notation,
3. statevector index,
4. measured classical-bit string.

These conventions can look different while representing the same physical state.

What Is Entanglement?

A two-qubit state is a **product state** if it can be written as

$$\boxed{|\psi\rangle_{AB} = |\phi\rangle_A \otimes |\chi\rangle_B}$$

for some single-qubit states $|\phi\rangle$ and $|\chi\rangle$.
If this factorization is impossible, the state is **entangled**.

Example of a product state:

$$\boxed{\frac{|00\rangle + |10\rangle}{\sqrt{2}} = |+\rangle \otimes |0\rangle}$$

Example of an entangled state:

$$\boxed{|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}}$$

Product State vs Entangled State

Product state

$$|\psi\rangle = |\phi\rangle_A \otimes |\chi\rangle_B$$

Example:

$$\frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

The state can be separated into two single-qubit states.

Entangled state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

There are no single-qubit states whose tensor product gives this state.

Entanglement is a property of the joint state.

Measurement of a Bell State

Consider

$$\left|\Phi^{+}\right\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Measurement probabilities:

$$P(00) = \frac{1}{2}, \qquad P(11) = \frac{1}{2}.$$

But

$$P(01) = P(10) = 0.$$

Therefore, the two measurement results are perfectly correlated. If the first qubit is measured as: 0, the second qubit is also:0. If the first qubit is measured as: 1, the second qubit is also:1.

Important

Entanglement is stronger than simply having correlated classical random variables. The joint quantum state cannot be decomposed into a tensor product of subsystem states.

How Do We Create Entanglement?

A standard recipe is

$$\boxed{\text{Superposition} + \text{Entangling gate} = \text{Entanglement}}$$

For example:

$$|00\rangle$$

First apply H to the first qubit:

$$|00\rangle \to \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle).$$

Then apply CNOT:

$$\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \to \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Thus,

$$\boxed{|00\rangle \xrightarrow{H} \xrightarrow{\text{CNOT}} |\Phi^+\rangle}$$

One-Minute Challenge

Consider

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle).$$

Questions

1. Is this state normalized?
2. What is the probability of measuring 01?
3. What is the probability of measuring 11?
4. Can the state be written as a product state?

One-Minute Challenge

Consider

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle).$$

Questions

1. Is this state normalized?
2. What is the probability of measuring 01?
3. What is the probability of measuring 11?
4. Can the state be written as a product state?

Since

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |1\rangle\,,$$

we have

$$\boxed{|\psi\rangle = |+\rangle \otimes |1\rangle}$$

so it is a product state.

Challenge: What Does CNOT Do?

Start with

$$|+0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle).$$

Apply CNOT.

Challenge: What Does CNOT Do?

Start with

$$|+0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle).$$

Apply CNOT.

$$\begin{aligned}\mathrm{CNOT}\,|+0\rangle &= \frac{1}{\sqrt{2}}\left(\mathrm{CNOT}\,|00\rangle + \mathrm{CNOT}\,|10\rangle\right) \\ &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).\end{aligned}$$

Therefore,

$$\boxed{|+0\rangle \xrightarrow{\mathrm{CNOT}} |\Phi^+\rangle}$$

A product state has become an entangled state.

Summary

Today we learned:

1. Multiple-qubit systems are described using tensor products:
$$\mathcal{H}_n = (\mathbb{C}^2)^{\otimes n}.$$
2. n qubits have 2^n computational-basis states.
3. Local operations act on individual qubits:
$$U \otimes I.$$
4. CNOT is a two-qubit controlled operation:
$$|c, t\rangle \to |c, c \oplus t\rangle .$$
5. Superposition followed by CNOT can create entanglement.
6. Qiskit uses q_0 as the least-significant bit for integer interpretation.
7. An entangled state cannot be factorized into a tensor product of individual subsystem states.

Take-Home Question

Consider the circuit

$$|00\rangle \xrightarrow{H\otimes I} \xrightarrow{\text{CNOT}} \left|\Phi^+\right\rangle .$$

Answer the following:

1. Write the state after the Hadamard gate.
2. Write the state after the CNOT gate.
3. What are the probabilities of $00, 01, 10, 11$?
4. Is the final state a product state or an entangled state?
5. Implement the circuit in Qiskit and verify the result experimentally using measurement shots.

Next step

Once we understand multiple qubits, we can construct larger quantum circuits and study multi-qubit algorithms.

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Thank You

Questions?

Multiple qubits → Tensor products → CNOT → Entanglement

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